Lagrange Duality (Part II)

Dual lower bounds, strong duality and KKT

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Another rare case: primal infeasible, dual finite

Consider

$$\min \ e^x \qquad \text{s.t.} \qquad e^x = 0.$$

Primal status

This problem is infeasible because $e^x > 0$ for all $x \in \mathbb{R}$. So

$$p^\star = +\infty.$$

The Lagrangian is

$$\mathcal{L}(x, \mu) = e^x + \mu e^x = (1 + \mu)e^x.$$

Now minimize over x:

$$g(\mu) = \inf_x (1 + \mu)e^x = \begin{cases} 0, & \mu \geq -1, \\ -\infty, & \mu < -1. \end{cases}$$

Therefore the dual problem is

$$\max_{\mu \geq -1} \ 0,$$

so

$$d^\star = 0.$$

Lesson

It is possible for the primal to be infeasible while the dual still has a finite optimum.

Strong duality

Definition

If

$$p^\star = d^\star$$

(and both are finite), then we say that **strong duality** holds.

- Under strong duality, the best lower bound produced by the dual is *exact*.
- This is the ideal situation: the dual completely certifies the primal optimum.

In linear programming

Strong duality always holds for LPs.
But for general optimization problems, strong duality may fail.

So the next question is:

Under what assumptions do we get $p^\star = d^\star$?

Why strong duality can fail

Strong duality can fail for several different reasons.

1. **Nonconvexity:** the dual lower bound may not be able to capture the true geometry of the primal problem.
2. **Degenerate feasible sets:** the feasible set may have empty interior or collapse to a lower-dimensional object.
3. **Constraint qualification failure:** even in convex problems, badly behaved constraints can prevent the dual from becoming tight.

Three examples we will discuss

- an SDP with a positive duality gap,
- a discrete-domain example,
- a convex problem whose constraints violate Slater's condition.

Failure example: a semidefinite program with a gap

Consider

$$\min \ y_1 \qquad \text{s.t.} \qquad \begin{pmatrix} 0 & y_1 & 0 \\ y_1 & y_2 & 0 \\ 0 & 0 & y_1+1 \end{pmatrix} \succeq 0.$$

Why must $y_1 = 0$?

For a PSD matrix, every diagonal entry must be nonnegative, and if a diagonal entry is 0, then the corresponding row/column cannot create a negative quadratic form.
Here the upper-left entry is 0. Consider the vector $(1, t, 0)^\top$:

$$(1, t, 0) \begin{pmatrix} 0 & y_1 & 0 \\ y_1 & y_2 & 0 \\ 0 & 0 & y_1+1 \end{pmatrix} \begin{pmatrix} 1 \\ t \\ 0 \end{pmatrix} = 2ty_1 + t^2 y_2.$$

For this to be nonnegative for every t, we must have $y_1 = 0$.

Then feasibility reduces to $y_2 \geq 0$, so

$$p^\star = 0.$$

But the dual value turns out to be $d^\star = -1$, giving a gap of 1.

Failure example 2: discrete domain

Let

$$D = [0,1], \qquad f(x) = \begin{cases} 1, & x = 0, \\ 0, & x \in (0,1]. \end{cases}$$

Consider

$$\min \; f(x) \qquad \text{s.t.} \qquad x \le 0, \quad x \in D.$$

Primal value

The constraints force $x = 0$, so

$$p^\star = f(0) = 1.$$

The Lagrangian is

$$\mathcal{L}(x,\lambda) = f(x) + \lambda x, \qquad \lambda \ge 0.$$

Now compute the dual function:

$$g(\lambda) = \inf_{x \in [0,1]} (f(x) + \lambda x).$$

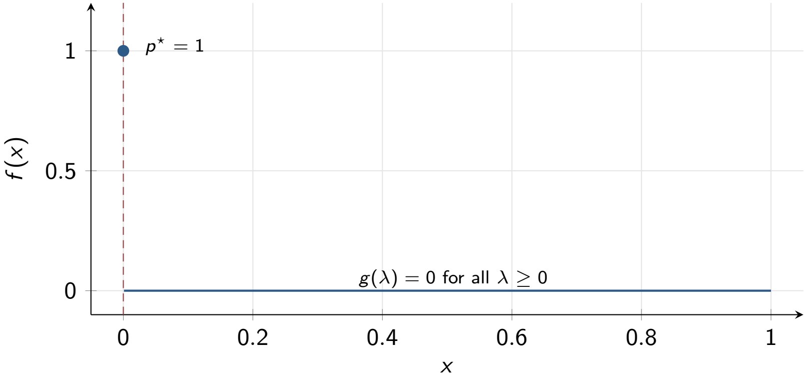
At $x = 0$, the value is 1. For any $x \in (0,1]$, the value is λx, whose infimum over $(0,1]$ is 0. Hence

$$g(\lambda) = 0 \qquad \text{for every } \lambda \ge 0.$$

Therefore

$$d^\star = 0 < 1 = p^\star.$$

Discrete-domain gap in a picture



Why the gap appears

The dual relaxation only sees the lower envelope produced by affine penalties. It cannot encode the jump discontinuity that forces the primal to stay at the isolated point $x = 0$.

Failure example 3: constraint qualification failure

Consider

$$\min \; e^{-x} \qquad \text{s.t.} \qquad \frac{x^2}{y} \le 0, \qquad y > 0.$$

Primal value

Because $y > 0$, the constraint $\frac{x^2}{y} \le 0$ forces

$$x^2 \le 0 \implies x = 0.$$

Any $y > 0$ is then feasible, so

$$p^\star = e^0 = 1.$$

The Lagrangian is

$$\mathcal{L}(x, y, \lambda) = e^{-x} + \lambda \frac{x^2}{y}, \qquad \lambda \ge 0.$$

Now choose $y = x^3$ and let $x \to +\infty$. Then

$$e^{-x} + \lambda \frac{x^2}{x^3} = e^{-x} + \frac{\lambda}{x} \to 0.$$

Hence

$$g(\lambda) = 0 \quad \forall \lambda \ge 0, \qquad d^\star = 0.$$

So there is a positive duality gap:

$$p^\star - d^\star = 1.$$

Slater's condition

Now we state the most important sufficient condition for strong duality in convex optimization.

Convex primal problem

$$\min \; f(x) \qquad \text{s.t.} \qquad f_i(x) \leq 0, \; i = 1, \ldots, m, \qquad Ax = b,$$

where f and each f_i are convex.

Slater's condition

There exists some $\bar{x} \in \text{int}(\mathcal{D})$ such that

$$f_i(\bar{x}) < 0 \quad \forall i, \qquad A\bar{x} = b.$$

In words: the inequality constraints can be satisfied **strictly**, while the equalities are satisfied exactly.

Nickname

Slater's condition is often called **strict feasibility**.

Why Slater's condition is a natural assumption

Slater's condition says the feasible set has **room to breathe**.

What it rules out

- feasible sets consisting only of boundary points,
- feasible sets with empty interior relative to the inequality constraints,
- pathological situations where the supporting hyperplane at the optimum becomes vertical.

Why it is often easy to satisfy

If your constraints are ordinary convex inequalities like

$$x > 0, \quad \|x\|_2 < 1, \quad a_i^\top x < b_i,$$

then checking strict feasibility is often straightforward.

But note

Slater's condition is about *inequalities*. Equality constraints are always satisfied exactly, never strictly.

Slater's theorem

Theorem (Slater)

For a convex optimization problem, if

- the optimal value $p^\star$ is finite, and
- Slater's condition holds,

then

$$p^\star = d^\star.$$

Moreover, the dual optimum is attained by some $(\lambda^\star, \mu^\star)$.

Interpretation

Under strict feasibility, the dual lower bound is not merely valid — it is **exact**.

This is the gateway to KKT

Once strong duality holds, primal and dual optima become tightly linked. The KKT conditions will express that linkage algebraically.

Why Slater implies strong duality: geometric intuition

For convex problems, define the epigraph-style set

$$\mathcal{A} = \{(u, t) : \exists x \in \mathcal{D} \text{ with } f_i(x) \le u_i,\ Ax = b,\ f(x) \le t\}.$$

- The primal optimum $p^\star$ is the smallest t with $(0, t) \in \mathcal{A}$.
- The dual problem searches for supporting hyperplanes under $\mathcal{A}$.

What can go wrong without Slater?

The supporting hyperplane at $(0, p^\star)$ might be **vertical**. A vertical hyperplane does not correspond to a useful dual multiplier vector.

What Slater prevents

If there is a strictly feasible point, then $\mathcal{A}$ extends into the region $u < 0$. That makes a vertical supporting hyperplane impossible, forcing a nonvertical hyperplane and hence a tight dual certificate.

The set $\mathcal{G}$ and the primal problem

Define

$$\mathcal{G} = \left\{ \big(f_1(x), \dots, f_m(x),\ h_1(x), \dots, h_p(x),\ f(x) \big) :\ x \in \mathcal{D} \right\}.$$

Write a typical point as (u, v, t), where

$$u \in \mathbb{R}^m, \qquad v \in \mathbb{R}^p, \qquad t \in \mathbb{R}.$$

Then:

- u records the inequality-constraint values,
- v records the equality-constraint values,
- t records the objective value.

Primal value from the geometry

The primal value is

$$p^\star = \inf\{t :\ (u, v, t) \in \mathcal{G},\ u \leq 0,\ v = 0\}.$$

So we look at the slice of $\mathcal{G}$ satisfying the constraints and read off the lowest possible t-coordinate.

The dual function as a supporting hyperplane

For any $\lambda \geq 0$ and $\mu \in \mathbb{R}^p$,

$$\mathcal{L}(x, \lambda, \mu) = f(x) + \lambda^\top u + \mu^\top v = (\lambda, \mu, 1)^\top (u, v, t).$$

So minimizing the Lagrangian over x means

$$g(\lambda, \mu) = \inf_{(u,v,t) \in \mathcal{G}} (\lambda, \mu, 1)^\top (u, v, t).$$

Geometric meaning

For each normal vector $(\lambda, \mu, 1)$, the dual function gives the lowest hyperplane of that slope that still lies below $\mathcal{G}$.

Why this is a lower bound

The hyperplane lies below $\mathcal{G}$, so its intercept at the feasible slice must be below the primal optimum. That is weak duality in geometric language.

The epigraph set $\mathcal{A}$

For convex problems, a more natural set is

$$\mathcal{A} = \{(u, v, t) : \exists x \in \mathcal{D} \text{ such that } f_i(x) \leq u_i,\ \ h_j(x) = v_j,\ \ f(x) \leq t\}\,.$$

Why $\mathcal{A}$ is useful

- $\mathcal{A}$ is an "upper-right" enlargement of $\mathcal{G}$.
- If the primal is convex, then $\mathcal{A}$ is a **convex set**.
- The primal value is

$$p^\star = \inf\{t :\ (0, 0, t) \in \mathcal{A}\}.$$

Dual interpretation

The dual function can also be viewed as the supporting hyperplane value of $\mathcal{A}$ with normal vector $(\lambda, \mu, 1)$.

Weak duality and strong duality from the picture

Weak duality

Every supporting hyperplane to $\mathcal{A}$ lies below the set. So its intercept at $(u, v) = (0, 0)$ must satisfy

$$g(\lambda, \mu) \leq p^\star.$$

Strong duality

If there exists a **nonvertical** supporting hyperplane to $\mathcal{A}$ at the point $(0, 0, p^\star)$, then its intercept equals $p^\star$. That hyperplane corresponds to an optimal dual pair, so

$$d^\star = p^\star.$$

Role of Slater

Slater's condition forces the supporting hyperplane at $(0, 0, p^\star)$ to be nonvertical.

Proof strategy under Slater: one inequality constraint

To keep notation light, we prove strong duality for

$$\min_x \; f(x) \qquad \text{s.t.} \qquad f_1(x) \le 0, \quad x \in \mathcal{D},$$

where f and f_1 are convex and $\mathcal{D}$ is convex.
Define

$$\mathcal{A} = \{(u,t) : \exists x \in \mathcal{D} \text{ with } f_1(x) \le u,\; f(x) \le t\}.$$

Goal

Show that there exists some $\lambda^\star \ge 0$ such that

$$g(\lambda^\star) = p^\star.$$

Roadmap

1. Separate $\mathcal{A}$ from the ray below $(0, p^\star)$.
2. Analyze the separating hyperplane.
3. Show its t-coefficient cannot be zero (this is where Slater is used).
4. Normalize the hyperplane to obtain a dual multiplier.

Step 1: define the set below the optimum

Let

$$\mathcal{B} = \{(0, s) : s < p^\star\}.$$

This is the vertical ray strictly below the point $(0, p^\star)$ on the t-axis.

Claim

$$\mathcal{A} \cap \mathcal{B} = \varnothing.$$

Proof.

If $(0, s) \in \mathcal{A}$, then by definition there exists some $x \in \mathcal{D}$ such that

$$f_1(x) \leq 0, \qquad f(x) \leq s.$$

So x is primal-feasible, and its objective value is at most s.
If also $s < p^\star$, then we have found a feasible point with value strictly below the optimal value — impossible.
Therefore $(0, s)$ with $s < p^\star$ cannot belong to $\mathcal{A}$. $\square$

Step 2: apply the separating hyperplane theorem

We now know that:

- $\mathcal{A}$ is convex,
- $\mathcal{B}$ is convex,
- $\mathcal{A}$ and $\mathcal{B}$ are disjoint.

So the separating hyperplane theorem gives numbers $(\lambda, \nu) \neq (0,0)$ and $a \in \mathbb{R}$ such that

$$\lambda u + \nu t \geq a \qquad \forall (u,t) \in \mathcal{A},$$

and

$$\nu s \leq a \qquad \forall s < p^\star.$$

What this means

The line

$$\lambda u + \nu t = a$$

separates the feasible epigraph set $\mathcal{A}$ from the ray strictly below the optimum.

Step 3: show that $\lambda \geq 0$ and $\nu \geq 0$

Because $\mathcal{A}$ is an epigraph-style set, it is monotone in the northeast direction: if $(u, t) \in \mathcal{A}$, then every (u', t') with

$$u' \geq u, \qquad t' \geq t$$

also lies in $\mathcal{A}$.

Now suppose $\lambda < 0$. Then for any fixed $(u, t) \in \mathcal{A}$, we can increase u to $+\infty$, and

$$\lambda u + \nu t \to -\infty,$$

which would violate the inequality

$$\lambda u + \nu t \geq a.$$

So $\lambda < 0$ is impossible. Hence

$$\lambda \geq 0.$$

Similarly, if $\nu < 0$, increasing t to $+\infty$ would violate separation. So

$$\nu \geq 0.$$

Conclusion

The separating hyperplane has a nonnegative normal:

$$\lambda \geq 0, \qquad \nu \geq 0.$$

Step 4: compare the separating line to the primal optimum

From

$$\nu s \leq a \qquad \forall s < p^\star,$$

take $s \uparrow p^\star$ to obtain

$$\nu p^\star \leq a.$$

Now take any $x \in \mathcal{D}$. The point

$$(f_1(x), f(x))$$

belongs to $\mathcal{A}$, so separation gives

$$\lambda f_1(x) + \nu f(x) \geq a.$$

Combining with $\nu p^\star \leq a$ yields

$$\lambda f_1(x) + \nu f(x) \geq \nu p^\star \qquad \forall x \in \mathcal{D}.$$

This is the crucial inequality. Everything now depends on whether $\nu > 0$ or $\nu = 0$.

Step 5A: if $\nu > 0$, normalize and get the dual bound

Assume first that $\nu > 0$.
Divide the inequality

$$\lambda f_1(x) + \nu f(x) \geq \nu p^\star$$

by ν:

$$f(x) + \frac{\lambda}{\nu} f_1(x) \geq p^\star \qquad \forall x \in \mathcal{D}.$$

Define

$$\lambda^\star = \frac{\lambda}{\nu} \geq 0.$$

Then

$$\mathcal{L}(x, \lambda^\star) = f(x) + \lambda^\star f_1(x) \geq p^\star \qquad \forall x \in \mathcal{D}.$$

Take the infimum over x:

$$g(\lambda^\star) = \inf_x \mathcal{L}(x, \lambda^\star) \geq p^\star.$$

But weak duality always gives

$$g(\lambda^\star) \leq p^\star.$$

Hence

$$g(\lambda^\star) = p^\star.$$

Done

We have found a dual-feasible point attaining the primal value, so strong duality holds.

Step 5B: if $\nu = 0$, Slater gives a contradiction

Now suppose instead that

$$\nu = 0.$$

Then the key inequality becomes

$$\lambda f_1(x) \geq 0 \qquad \forall x \in \mathcal{D}.$$

Since Slater's condition holds, there exists some strictly feasible $\bar{x}$ such that

$$f_1(\bar{x}) < 0.$$

Substitute $x = \bar{x}$:

$$\lambda f_1(\bar{x}) \geq 0.$$

But we already know $\lambda \geq 0$, and $f_1(\bar{x}) < 0$, so the only way

$$\lambda f_1(\bar{x}) \geq 0$$

can hold is if

$$\lambda = 0.$$

Thus

$$(\lambda, \nu) = (0, 0),$$

which contradicts the separating hyperplane theorem.

Conclusion

The case $\nu = 0$ is impossible. Therefore $\nu > 0$, and the previous slide applies.

What the proof really used

Let's summarize the logic.

1. Separation gives a supporting line at the optimum.
2. To turn that line into a dual multiplier, we need its t-coefficient ν to be positive.
3. Slater's condition forces exactly that by ruling out a vertical supporting line.

Conceptual takeaway

Strong duality is fundamentally a **geometric separation statement**.

Extension to many inequalities

For m inequality constraints, the same proof works in $\mathbb{R}^{m+1}$. The scalar λ is replaced by a vector $\lambda \in \mathbb{R}^m_{\geq 0}$, and the same normalization argument yields an optimal dual vector $\lambda^\star$.

From strong duality to complementary slackness

Suppose strong duality holds, and let $x^\star$ be primal optimal and $(\lambda^\star, \mu^\star)$ dual optimal. Then

$$f(x^\star) = p^\star = d^\star = g(\lambda^\star, \mu^\star).$$

Now compare this to the Lagrangian:

$$g(\lambda^\star, \mu^\star) = \inf_x \mathcal{L}(x, \lambda^\star, \mu^\star) \leq \mathcal{L}(x^\star, \lambda^\star, \mu^\star).$$

Because $x^\star$ is feasible,

$$\mathcal{L}(x^\star, \lambda^\star, \mu^\star) = f(x^\star) + \sum_i \lambda_i^\star f_i(x^\star) + \sum_j \mu_j^\star h_j(x^\star) \leq f(x^\star),$$

since $h_j(x^\star) = 0$ and $\lambda_i^\star f_i(x^\star) \leq 0$.
So we have a chain

$$f(x^\star) \leq \mathcal{L}(x^\star, \lambda^\star, \mu^\star) \leq f(x^\star),$$

hence everything is equal.

Complementary slackness

From the equality

$$\mathcal{L}(x^\star, \lambda^\star, \mu^\star) = f(x^\star),$$

we obtain

$$\sum_{i=1}^{m} \lambda_i^\star f_i(x^\star) = 0.$$

But every term in the sum is nonpositive:

$$\lambda_i^\star \geq 0, \qquad f_i(x^\star) \leq 0.$$

A sum of nonpositive numbers can equal 0 only if each term is individually zero. Therefore

$$\boxed{\lambda_i^\star f_i(x^\star) = 0 \qquad \text{for every } i.}$$

Interpretation

Either

- the constraint is **active**: $f_i(x^\star) = 0$, in which case $\lambda_i^\star$ may be positive; or
- the constraint is **inactive**: $f_i(x^\star) < 0$, in which case necessarily $\lambda_i^\star = 0$.

The KKT conditions

KKT system

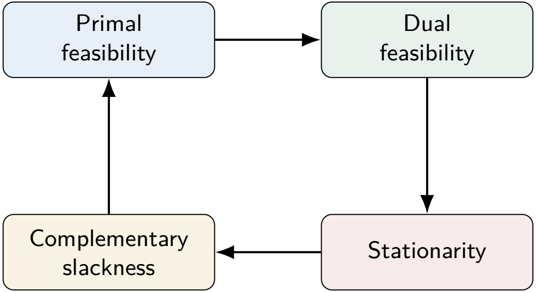
A triple (x, λ, μ) satisfies the Karush-Kuhn-Tucker conditions if:

1. **Primal feasibility:**
$$f_i(x) \leq 0, \qquad h_j(x) = 0.$$
2. **Dual feasibility:**
$$\lambda_i \geq 0.$$
3. **Complementary slackness:**
$$\lambda_i f_i(x) = 0.$$
4. **Stationarity:**
$$\nabla f(x) + \sum_i \lambda_i \nabla f_i(x) + \sum_j \mu_j \nabla h_j(x) = 0.$$

Mnemonic

feasible primal $+$ feasible dual $+$ no wasted multiplier $+$ zero gradient.

KKT conditions as a flow diagram



How to read this

At optimum, these four conditions must hold simultaneously. In convex problems with strong duality, this system is equivalent to optimality.

The KKT-optimality theorem

Theorem

For convex optimization problems:

1. If strong duality holds and $(x^\star, \lambda^\star, \mu^\star)$ are primal-dual optimal, then they satisfy KKT.
2. If a triple $(\bar{x}, \bar{\lambda}, \bar{\mu})$ satisfies KKT, then $\bar{x}$ is primal optimal and $(\bar{\lambda}, \bar{\mu})$ is dual optimal.
3. Therefore, under Slater's condition,

$$\bar{x} \text{ is primal optimal} \iff \exists(\bar{\lambda}, \bar{\mu}) \text{ such that KKT holds.}$$

We prove both directions carefully.

Proof that strong duality implies KKT

Suppose $x^\star$ and $(\lambda^\star, \mu^\star)$ are primal and dual optimal, and strong duality holds.
We already showed:

$$f(x^\star) = g(\lambda^\star, \mu^\star) = \inf_x \mathcal{L}(x, \lambda^\star, \mu^\star) = \mathcal{L}(x^\star, \lambda^\star, \mu^\star).$$

This gives two facts immediately:

1. Since $x^\star$ attains the infimum of the Lagrangian,

   $$x^\star \in \arg\min_x \mathcal{L}(x, \lambda^\star, \mu^\star).$$

   Because the problem is differentiable,

   $$\nabla_x \mathcal{L}(x^\star, \lambda^\star, \mu^\star) = 0.$$

   That is stationarity.

2. Since

   $$\mathcal{L}(x^\star, \lambda^\star, \mu^\star) = f(x^\star),$$

   we get

   $$\sum_i \lambda_i^\star f_i(x^\star) = 0,$$

   which yields complementary slackness term by term.

Primal feasibility holds because $x^\star$ is primal optimal; dual feasibility holds because $(\lambda^\star, \mu^\star)$ is dual optimal.

Proof that KKT implies optimality: the main idea

Now suppose $(\bar{x}, \bar{\lambda}, \bar{\mu})$ satisfies KKT.
We want to prove that

$$f(\bar{x}) = p^\star = d^\star = g(\bar{\lambda}, \bar{\mu}).$$

The proof has two ingredients:

1. Stationarity says that $\bar{x}$ minimizes the Lagrangian $\mathcal{L}(x, \bar{\lambda}, \bar{\mu})$.
2. Complementary slackness and primal feasibility say that, at $\bar{x}$, the Lagrangian equals the primal objective.

If both statements hold, then the dual function evaluated at $(\bar{\lambda}, \bar{\mu})$ exactly equals the primal value at $\bar{x}$.
Weak duality will then force optimality.

KKT implies optimality: step-by-step proof

Because the problem is convex and $\bar{\lambda}_i \geq 0$, the Lagrangian

$$x \mapsto \mathcal{L}(x, \bar{\lambda}, \bar{\mu}) = f(x) + \sum_i \bar{\lambda}_i f_i(x) + \sum_j \bar{\mu}_j h_j(x)$$

is a convex function of x.

By stationarity,

$$\nabla_x \mathcal{L}(\bar{x}, \bar{\lambda}, \bar{\mu}) = 0.$$

For a differentiable convex function, zero gradient means global minimizer. Hence

$$g(\bar{\lambda}, \bar{\mu}) = \inf_x \mathcal{L}(x, \bar{\lambda}, \bar{\mu}) = \mathcal{L}(\bar{x}, \bar{\lambda}, \bar{\mu}).$$

Now expand:

$$\mathcal{L}(\bar{x}, \bar{\lambda}, \bar{\mu}) = f(\bar{x}) + \sum_i \bar{\lambda}_i f_i(\bar{x}) + \sum_j \bar{\mu}_j h_j(\bar{x}).$$

By primal feasibility,

$$h_j(\bar{x}) = 0.$$

By complementary slackness,

$$\bar{\lambda}_i f_i(\bar{x}) = 0.$$

So

$$g(\bar{\lambda}, \bar{\mu}) = f(\bar{x}).$$

KKT implies optimality: finishing the proof

We have just shown

$$g(\bar{\lambda}, \bar{\mu}) = f(\bar{x}).$$

Now compare this to primal and dual optima.

- Since $\bar{x}$ is primal-feasible,

$$f(\bar{x}) \geq p^\star.$$

- Since $(\bar{\lambda}, \bar{\mu})$ is dual-feasible,

$$g(\bar{\lambda}, \bar{\mu}) \leq d^\star.$$

- By weak duality,

$$d^\star \leq p^\star.$$

Putting these together:

$$p^\star \leq f(\bar{x}) = g(\bar{\lambda}, \bar{\mu}) \leq d^\star \leq p^\star.$$

Therefore all inequalities are equalities:

$$f(\bar{x}) = p^\star = d^\star = g(\bar{\lambda}, \bar{\mu}).$$

Conclusion

A KKT point is automatically a primal-dual optimal pair.

Under Slater: KKT $\iff$ optimality

Now combine the last theorem with Slater's theorem.

- Slater $\implies$ strong duality.
- Strong duality $\implies$ every primal-dual optimum satisfies KKT.
- KKT $\implies$ primal-dual optimality.

So for convex problems satisfying Slater's condition:

$$x^\star \text{ is optimal } \iff \exists(\lambda^\star, \mu^\star) \text{ such that KKT holds.}$$

Why this is important computationally

Instead of directly solving the constrained optimization problem, we can often solve the system

$$\begin{cases} f_i(x) \le 0, \\ h_j(x) = 0, \\ \lambda_i \ge 0, \\ \lambda_i f_i(x) = 0, \\ \nabla f(x) + \sum_i \lambda_i \nabla f_i(x) + \sum_j \mu_j \nabla h_j(x) = 0. \end{cases}$$

Summary of the chapter

- The **Lagrangian**

$$\mathcal{L}(x,\lambda,\mu) = f(x) + \sum_i \lambda_i f_i(x) + \sum_j \mu_j h_j(x)$$

  converts hard constraints into priced violations.
- The **dual function**

$$g(\lambda,\mu) = \inf_x \mathcal{L}(x,\lambda,\mu)$$

  is always concave and always provides a lower bound on the primal optimum.
- **Weak duality** always holds:

$$d^\star \le p^\star.$$

- **Strong duality** does not always hold, but for convex problems it is guaranteed by **Slater's condition**.
- Under strong duality, the **KKT conditions** characterize optimal primal-dual pairs.
- The optimal multiplier $\lambda^\star$ is also a **shadow price**: it measures the sensitivity of the optimal value to constraint relaxation.